

INVENTORY ANALYTICS USING POWER BI

A Practical Dashboard Approach

Portfolio Project Report

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Project focus: Inventory monitoring, stock movement, supplier analysis, and interactive business reporting

ABSTRACT This report presents a Power BI inventory analytics dashboard designed to organize and visualize stock information for business monitoring. The project uses key performance indicators, product- and supplier-level analysis, stock-out trends, and interactive filters to make inventory information easier to interpret. The report focuses on dashboard structure, analytical purpose, and business usefulness rather than claiming results that are not directly supported by the project data.

Keywords: Power BI | Inventory Analytics | Data Visualization | Stock Management | Business Intelligence

1. Introduction

Inventory management requires clear visibility into stock availability, incoming quantities, outgoing quantities, low-stock situations, product-level balances, and supplier contribution. When inventory information is stored only as rows in a spreadsheet or raw table, important patterns can be difficult to identify quickly. Business intelligence tools such as Microsoft Power BI help convert structured inventory data into interactive visuals and summary indicators that support faster interpretation.

This project was developed as a practical inventory analytics dashboard. The Power BI report contains a set of key performance indicator cards and analytical visuals connected to a raw data table. The design allows a user to move from an overall inventory view to product, supplier, and time-based analysis.

2. Project Objectives

- Present important inventory indicators in a clear dashboard format.
- Track total stock, total stock-in, total stock-out, and low-stock count.
- Compare closing stock across different products.
- Analyze how closing stock is distributed across suppliers.
- Observe stock-out movement over time using sales-date information.
- Allow users to filter the dashboard by supplier and product.
- Demonstrate practical Power BI skills in business reporting and inventory analytics.

3. Tools and Data Structure

The project was created in Microsoft Power BI. The report model uses a table named "Raw data" and includes measures and fields used across the dashboard. The visible report configuration references the following inventory elements:

Data element	Type / use	Purpose in the dashboard
Total Stock	Measure / KPI	Overall inventory quantity
Total Stock In	Measure / KPI	Incoming stock quantity
Total Stock Out	Measure / KPI	Outgoing stock quantity
Low Stock Count	Measure / KPI	Number of low-stock items
Product Name	Category field	Product-level filtering and comparison
Closing Stock	Numeric field	Remaining stock analysis
Supplier Name	Category field	Supplier-level analysis and filtering
Sales Date	Date field	Time-based stock-out trend
Stock Out	Numeric field	Quantity used in stock-out trend analysis

4. Dashboard Components

4.1 Key Performance Indicator Cards

The first dashboard view contains four card visuals: Total Stock, Total Stock In, Total Stock Out, and Low Stock Count. These cards provide a quick summary of inventory status before the user explores detailed charts. KPI cards are useful because they reduce the time required to locate the most important inventory measures.

4.2 Closing Stock by Product

A clustered column chart compares Product Name against the sum of Closing Stock. The chart is sorted by closing stock, helping users identify products with relatively higher or lower remaining inventory. This visual supports product-level inventory monitoring and can assist with replenishment discussions when combined with business rules and demand information.

4.3 Closing Stock by Supplier

A pie chart uses Supplier Name as the category and Closing Stock as the value. This view provides a supplier-level perspective of the inventory represented in the dataset. It can be used to understand how stock is distributed among suppliers and to support supplier-focused review.

4.4 Stock-Out Trend Over Time

A line chart plots Sales Date against Stock Out. The date hierarchy supports time-based analysis across year, quarter, month, and day levels. This allows the user to examine changes in outgoing stock over time and identify periods that may deserve further investigation.

4.5 Interactive Filters and Q&A

The report includes slicers for Supplier Name and Product Name, allowing users to focus the dashboard on selected categories. A Power BI Q&A visual is also present, providing an additional way to explore the model through natural-language questions where the underlying model supports them.

5. Analytical Workflow

The dashboard follows a simple business-intelligence workflow. Raw inventory data is structured in Power BI, measures summarize important inventory quantities, and visuals convert those measures and fields into decision-oriented views. The user can begin with overall KPIs, move to product and supplier comparisons, review stock movement over time, and then apply filters for more focused analysis.

Raw Inventory Data	Power BI Model	KPI Measures	Charts & Filters	Business Review
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6. Business Value of the Dashboard

- Faster inventory overview: KPI cards summarize core stock measures at a glance.
- Product visibility: closing-stock comparison helps users review inventory at product level.
- Supplier visibility: supplier-based analysis supports understanding of stock distribution by supplier.
- Trend analysis: the stock-out line chart helps users examine movement across time periods.
- Interactive exploration: supplier and product slicers allow users to narrow the analysis without changing the underlying report.
- Communication: visual reporting can make inventory information easier to explain to non-technical users.

7. Skills Demonstrated

The project demonstrates practical use of Microsoft Power BI for business reporting and inventory analysis. It also demonstrates the ability to organize data elements around a business problem and select visuals that match the purpose of each measure or category.

- Microsoft Power BI dashboard development
- Inventory and stock-movement analysis
- Data visualization and KPI design
- Product- and supplier-level reporting
- Time-series visualization using date hierarchies
- Interactive slicers and report filtering
- Business data interpretation and presentation

8. Limitations and Opportunities for Improvement

This report documents the current dashboard structure and analytical design. It does not claim specific business outcomes or numerical findings because the project file, as reviewed for this report, is being used primarily to demonstrate dashboard construction and inventory-analysis capability. Future versions could extend the project with additional business measures and validation rules.

- Add reorder-level and safety-stock measures to support replenishment decisions.
- Create inventory turnover and days-of-inventory metrics when the required cost and demand data are available.
- Add conditional alerts for low-stock or zero-stock items.

- Include supplier performance measures such as lead time, delivery reliability, and purchase quantities when those fields are available.
- Add drill-through pages for individual products or suppliers.
- Create a mobile-optimized report layout for easier viewing on smaller screens.

9. Conclusion

The Inventory Analytics Dashboard demonstrates how Power BI can convert inventory data into a structured and interactive business report. By combining KPI cards, product and supplier comparisons, stock-out trends, and slicers, the dashboard provides multiple levels of inventory visibility within a single project. The project is suitable as a portfolio example because it connects technical dashboard-building skills with a practical business use case.

The main learning outcome is that an effective dashboard is not only a collection of charts. Each visual should answer a clear business question: how much stock is available, how stock moves in and out, which products hold the remaining inventory, how suppliers contribute to stock, and how outgoing stock changes over time.

10. Portfolio Publication Note

This document is a portfolio project report prepared to accompany the Power BI Inventory Analytics project. It may be uploaded to a GitHub portfolio or shared through professional networking platforms as evidence of project-based learning. It should be described as a project report or technical article rather than as a peer-reviewed academic publication unless it is later formally published by an academic or professional venue.

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